

# Vibe-IC — All Steps: Phase → Stage → Step

Every step of the full flow, organised as **Phase → Stage → Step** with a **single continuous numbering 1 → 44** (starting from Stage 1's Spec-to-RTL). Phase 1's document-generation steps are labelled **D1–D5** (pre-flow, not counted in 1→44); two parallel tracks — **Analog A1–A9** and **Mixed-signal M1–M4** — run alongside.

## Phase → Stage map

- **Phase 1** — Specification & documents: two entries (**Agent path** • **doc-gen path D1–D5**) + optional architecture-exploration front-ends
- **Phase 2** — RTL → synthesis: Stage 1 (RTL+verification) • Stage 2 (constraints+synthesis+DFT+handoff gate)
- **Phase 3** — Physical → Tapeout: Stage 3 (physical+sign-off) • Stage 4 (output+tapeout) • Stage 5 (manufacturing & test)
- **Parallel** — Analog A1–A9 • Mixed-signal M1–M4

## Phase 1 — Specification & documents

Two entries; both produce the same L-series design documents that feed Phase 2:

- phase1/input\_prompt/ — free text / natural language → **Agent path**
- phase1/input\_doc/ — existing documents / structured YAML → **doc-gen path** (D1–D5)

### Agent path (free-text input)

| Agent           | What it does                                                                                                                            |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| PM Agent        | User-facing: turns natural-language requirements into design facts, asks one plain-language question per gap, hands off once confirmed. |
| IC Expert Agent | Behind the PM: reviews every layer with silicon expertise, fills sensible defaults, cross-layer consistency checks. Never user-facing.  |

Flow: user free text → PM Agent → IC Expert Agent → finalised L-series documents.

### doc-gen path D1–D5 (existing-document input)

| #  | Step                          | What it does                                                                             | Input               | Output                    | Tools (EDA)              | Programs / Skills                                                                                                                                                                                                                                                                                                                                                                                                  |
|----|-------------------------------|------------------------------------------------------------------------------------------|---------------------|---------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D1 | Doc extraction → L1–L13       | Ingest prompt/docs into input_doc/ and deterministically extract the core design layers. | user docs / prompt  | L1–L13 JSON               | deterministic extractors | phase1_all_l_docs_present_check<br>skills: datasheet-gen • frs-gen • cmd-protocol-gen • regmap-gen • ad<br>i-spec-gen • control-logic-gen • t<br>est-debug-gen • timing-waveform-<br>gen • rtl-constants-gen • integrat<br>ion-spec-gen • test-cases-gen • ca<br>libration-gen • behavioral-<br>sequences-gen • lab-calibration-<br>gen • otp-content-gen • doc-<br>consistency-check • schematic-<br>gen • phase1 |
| D2 | Core design-layer docs L1–L13 | Deterministic extraction of datasheet, FRS, register map, etc.                           | D1 plain text       | L1_DATASHEET ... L13      | deterministic extractors | —                                                                                                                                                                                                                                                                                                                                                                                                                  |
| D3 | Extended docs L14–L23         | Protocol, timing, power intent (L21), skeletons.                                         | L1–L13              | L14–L23 JSON              | overlay extractor        | —                                                                                                                                                                                                                                                                                                                                                                                                                  |
| D4 | Protocol-class synthesis      | Detect the IC's protocol class (81 classes) and synthesise class facts.                  | full input text     | ic_class + protocol facts | is_+_synth               | —                                                                                                                                                                                                                                                                                                                                                                                                                  |
| D5 | Coverage report               |                                                                                          | input docs + L docs | parity / coverage report  | parity reporter          | —                                                                                                                                                                                                                                                                                                                                                                                                                  |

| # | Step | What it does                                                | Input | Output | Tools (EDA) | Programs / Skills |
|---|------|-------------------------------------------------------------|-------|--------|-------------|-------------------|
|   |      | Verify the input documents landed completely in the L docs. |       |        |             |                   |

### Architecture-exploration front-ends (optional, feed Step 1)

| Front-end                  | What it does                                                                            | Input                 | Output                               |
|----------------------------|-----------------------------------------------------------------------------------------|-----------------------|--------------------------------------|
| architecture-explore       | Design-space exploration (pipeline depth / parallelism / memory vs PPA, Pareto filter). | L docs + perf targets | architecture decisions (feed Step 1) |
| hls-c2rtl                  | C/C++/SystemC → RTL via HLS (open-source XLS, etc.).                                    | C/SystemC model       | RTL (verified from Step 2 onward)    |
| SpinalHDL/Chisel front-end | eda_spinalhdl_gen emits Verilog from Scala HDL.                                         | SpinalHDL source      | Verilog RTL                          |

Front-end precedence (**artifact-driven**): existing RTL > C/SystemC model (hls-c2rtl) > SpinalHDL/Chisel > prompt-only (Step 1 spec-to-RTL). The available artifacts decide the path — never pick one speculatively.

## Phase 2 — RTL → synthesis

### Stage 1 — RTL generation & verification

| # | Step                                                                                                                  | What it does                                                                                                                                                                                                                                          | Input                | Output                                         | Tools (EDA)                                     | Programs / Skills                                                                                                                                     |
|---|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|------------------------------------------------|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Spec-to-RTL                                                                                                           | Author synthesizable RTL from the L-series docs (SoC/CPU classes may take the IP-catalog reuse + glue path, e.g. Caravel-class harness platforms).                                                                                                    | L1–L23 docs          | rtl/*.v(.sv) • coverage report                 | — (AI-authored from L docs; SoC via IP-catalog) | skills: spec-to-rtl                                                                                                                                   |
| 2 |  Lint (RTL hygiene)                  | Static RTL style/bug checks; auto-fixable issues fixed first.                                                                                                                                                                                         | RTL                  | lint reports (hygiene / ROM-init)              | Verilator lint                                  | rtl_hygiene_lint • rom_init_lint • rtl_bug_report_schema_check • internal_vs_external_timing_check...<br>skills: rtl-review                           |
| 3 |  CDC / RDC check                   | Clock-domain / reset-domain crossing safety.                                                                                                                                                                                                          | RTL                  | CDC/RDC reports (crossing / async / reset-dep) | in-house CDC/RDC scan                           | cdc_crossing_check • cdc_async_input_check • reset_dependency_check...<br>skills: cdc-check • rdc-check                                               |
| 4 |  Simulation (testbench + coverage) | Per-IC oracle testbench functional simulation (golden compares) + coverage measurement; when L21 declares power domains, the TB should cover power-state transition scenarios (no open-source UPF-aware sim — structural verification stays with M2). | RTL • L10 test cases | sim logs • results.xml • coverage report       | iverilog/vvp • Verilator coverage eda_simulate  | testbench_gen • coverage_closure • l10_tb_conformance_check • l12_tb_coverage_check...<br>skills: testbench-gen                                       |
| 5 |  Formal verification (assertions)  | Prove key properties hold: safety invariants proved unbounded and functional properties bounded-model-checked with the bound disclosed (a1_l_proved requires the .sby + SymbiYosys evidence chain).                                                   | RTL • L3 constraints | .sby • formal results • full-stack TB results  | SymbiYosys (ABC pdr / bmc3) eda_formal          | formal_property_run • assertion_property_check • bit_level_full_stack_tb_check • formal_proof_evidence_check<br>skills: assertion-gen • formal-verify |

| # | Step                 | What it does                                | Input                   | Output                                      | Tools (EDA)          | Programs / Skills                                                                                                           |
|---|----------------------|---------------------------------------------|-------------------------|---------------------------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 6 | FPGA early prototype | Pre-synthesis behavioral prototype on FPGA. | RTL • board constraints | .sof • map report • FPGA verification audit | Quartus<br>eda_synth | fpga_test_harness_gen • debug_fir<br>st_pass • quartus_map_audit • fpga<br>_verification_audit<br>skills: fpga-test-harness |

## Stage 2 — Constraints, synthesis, DFT & handoff gate

| #   | Step                                                                                                                         | What it does                                                                                                                                                                                                                                                                                                                                                | Input                                  | Output                                                                          | Tools (EDA)                                                      | Programs / Skills                                                                                                                            |
|-----|------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 7   | Constraint setup (SDC + PVT matrix)                                                                                          | Author timing constraints (SDC) + PVT corner matrix; power intent modelled via L21 and rendered to UPF (handoff artifact — open-source tools do not consume UPF; structural verification stays with M2).                                                                                                                                                    | L8 timing • L21 • PDK liberty          | *.sdc • pvt_matrix.json • <top>.upf (optional, when L21 declares power domains) | — (UPF emitted by the l21_to_upf_emit script, not an EDA engine) | sdcsyntaxcheck • pvt_matrix_cke<br>ck • l21_to_upf_emit • upf_syntax_<br>check<br>skills: constraint-gen                                     |
| 8   |  SDC validation (incl. derived-clock guard) | Validate constraint correctness, completeness, and exception (false_path/multicycle) justification; manual ICG / register-divided clocks must declare a matching create_generated_clock (vacuous PASS without derived clocks — i.e. it passes automatically when the condition does not apply: no divided clocks in the design means the check self-skips). | SDC • L8 • RTL                         | SDC check report • derived_clock_sdc.json                                       | —                                                                | sdcsyntaxcheck • sdc_validator_<br>check •<br>sdc_exception_correlation_check •<br>derived_clock_sdc_required_check<br>skills: sdc-validator |
| 9   | Synthesis (Yosys → mapped netlist)                                                                                           | Synthesize RTL and technology-map (dfflibmap + abc -liberty) to the standard-cell netlist.                                                                                                                                                                                                                                                                  | RTL • SDC • liberty                    | synth/netlist.v • area stats                                                    | Yosys + abc<br>eda_synth                                         | synth_wrapper_gen • synth_netlist_<br>_check • provenance_check<br>skills: synth-doctor                                                      |
| 10  |  Pre-layout STA (multi-corner)              | Pre-layout static timing analysis (SS/TT/FF) + post-synth power preview (gate-level vectorless, default toggle rate, disclosed via analysis_mode; a different accuracy tier from Step 33's post-layout, optionally VCD-vector, analysis).                                                                                                                   | netlist • SDC • liberty                | pre-PnR timing report + summary • pre_pnr_power_preview.rpt                     | OpenSTA<br>eda_sta                                               | sta_report_check<br>skills: sta-review                                                                                                       |
| 11  | DFT insertion (scan chain + ATPG)                                                                                            | Insert scan chains + generate patterns (open-source Fault: scan + stuck-at ATPG + TAP; MBIST/LBIST/compression out of open-source scope).                                                                                                                                                                                                                   | netlist                                | scan netlist • ATPG coverage report                                             | Fault (scan+ATPG+TAP)<br>eda_dft                                 | fault_atpg_run • dft_atpg_coverag<br>e_check<br>skills: dft-insert • atpg                                                                    |
| 12  | Post-DFT optimization (resynth / buffering)                                                                                  | Re-optimize timing/area after DFT insertion.                                                                                                                                                                                                                                                                                                                | scan netlist                           | post_dft_netlist.v                                                              | Yosys resynth                                                    | skills: synth-doctor                                                                                                                         |
| FS1 | ISO-26262 FMEDA diagnostic-coverage (safety designs only)                                                                    | Fault-injection on a declared safety mechanism (ECC/parity): inject stuck-at faults on the protected path and measure diagnostic coverage against the ASIL floor. Not applicable for non-safety designs.                                                                                                                                                    | RTL • declared safety mechanism • ASIL | fmEDA_coverage report (measured DC)                                             | iverilog fault-injection                                         | fmEDA_fault_injection_coverage •<br>fmEDA_coverage_check                                                                                     |
| DT1 | Transition-delay-fault (at-speed LOC) ATPG                                                                                   | Generate launch-capture 2-patterns for transition faults on a 2-time-frame launch-on-capture unroll of the scan-cut netlist, and grade the transition test coverage. Not applicable for combinational / no-scan designs.                                                                                                                                    | scan-cut netlist • clock               | transition coverage report                                                      | Yosys SAT (sat -prove)                                           | transition_fault_atpg_run • trans<br>ition_coverage_check                                                                                    |
| DT2 | Path-delay-fault (at-speed, timing-graded) ATPG                                                                              | Report the K longest launch-on-capture paths from the routed netlist with real post-layout timing, then generate and grade a launch-capture                                                                                                                                                                                                                 | routed netlist • SPEF • SDC • scan cut | path-delay coverage report                                                      | OpenSTA + Yosys SAT (sat -prove)                                 | path_delay_fault_atpg_run • path_<br>delay_coverage_check                                                                                    |

| #   | Step                                                                                                                           | What it does                                                                                                                                                                                                                                                                                                                                                                                                          | Input                                           | Output                | Tools (EDA)                      | Programs / Skills                                          |
|-----|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|-----------------------|----------------------------------|------------------------------------------------------------|
|     |                                                                                                                                | 2-pattern per path (robust and non-robust); a path with no possible pattern is excluded, never counted. Not applicable before the routed netlist and parasitics exist.                                                                                                                                                                                                                                                |                                                 |                       |                                  |                                                            |
| DT3 | Small-delay-defect (SDD) at-speed grade                                                                                        | Grade each timing-critical path fault by its real post-layout slack: a defect is caught at-speed only when the detecting path's margin is tight, so detection through a low-slack path is a strong catch and through a slacky path is weak. A slack-rich design honestly scores low (its margin masks small delays). Descriptive grade, no floor. Not applicable before the path-delay and transition coverage exist. | path-delay coverage • transition coverage • SDC | SDD coverage report   | OpenSTA + Yosys SAT (sat -prove) | sdd_atpg_run • sdd_coverage_check                          |
| 13  |  Equivalence check (RTL ≡ netlist)            | Formally prove gate-level netlist ≡ RTL (LEC).                                                                                                                                                                                                                                                                                                                                                                        | RTL • post-DFT netlist                          | LEC report            | Yosys equiv                      | lec_equivalence_check<br>skills: equivalence-check         |
| 14  |  Synthesis handoff gate (pre-PnR Yosys audit) | Synthesis→PnR handoff QA: synth script + netlist audit ( <b>open-source-Yosys specific</b> ; the synthesis stage's closing gate).                                                                                                                                                                                                                                                                                     | synth script • netlist                          | handoff audit reports | Yosys script/netlist audit       | yosys_hilomap_required_check • yosys_script_template_check |

## Phase 3 — Physical design → Tapeout

### Stage 3 — Physical design & sign-off

| #  | Step                                                                                                     | What it does                                                                                                                     | Input                                                                                                                                                                                               | Output                             | Tools (EDA)                                      | Programs / Skills                                                 |
|----|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|--------------------------------------------------|-------------------------------------------------------------------|
| 15 | Floorplan + PDN                                                                                          | Chip floorplan + power delivery network; tapcell insertion (latch-up well-ties, SKY130 14 µm rule).                              | netlist • hardmacro LEFs (pdk_local/ auto-included, via the IP integration check: LEF/GDS/Liberty alignment + corner coverage + L21 supply consistency; macro LEFs should carry obstruction layers) | floorplan.def • PDN                | OpenROAD (init_floorplan+pdngen+tapcell) eda_pnr | phase3_backend_step • floorplan_pdn_check • ip_integration_check  |
| 16 | Clock planning                                                                                           | Clock-tree distribution strategy.                                                                                                | floorplan                                                                                                                                                                                           | clock_plan.json                    | OpenROAD CTS planning                            | clock_plan_check<br>skills: cts-plan                              |
| 17 | Placement (global + detailed)                                                                            | Place standard cells.                                                                                                            | floorplan • netlist                                                                                                                                                                                 | placed.def                         | OpenROAD (global+detailed place) eda_pnr         | placement_legality_check<br>skills: placement-optimize            |
| 18 | Spare-cell + ECO-prep insertion                                                                          | Pre-place spare cells / ECO readiness so later fixes are metal-only (paired with Step 32 ECO: provisioned here, consumed there). | placed.def                                                                                                                                                                                          | spare_cells.json • coverage report | OpenROAD eda_pnr                                 | spare_cell_coverage_check • spare_cell_preservation_check         |
| 19 | CTS (clock tree synthesis)                                                                               | Build and balance the clock tree.                                                                                                | placed.def • clock plan                                                                                                                                                                             | post_cts.def • clock-tree report   | OpenROAD CTS eda_pnr                             | cts_quality_check                                                 |
| 20 |  Post-CTS hold fixing | Repair hold violations after CTS (the runner repeats hold repair post-global-route).                                             | post_cts.def                                                                                                                                                                                        | post_hold.def                      | OpenROAD repair_timing -hold                     | hold_closure_check<br>skills: hold-fix                            |
| 21 | Routing (global + detailed)                                                                              | Complete all signal routing.                                                                                                     | post_hold.def                                                                                                                                                                                       | routed.def • router DRC report     | OpenROAD TritonRoute eda_pnr                     | drc_report_check • def_stage_progression_check • provenance_check |
| 22 | Parasitic extraction (SPEF)                                                                              | Extract post-route parasitics; when the PDK ships no coupling captable, lateral coupling caps are added                          | routed.def • tech LEF                                                                                                                                                                               | SPEF (coupling-aware)              | OpenRCX eda_extraction                           | spef_extraction_check • provenance_check                          |

| #  | Step                                                                                                                                          | What it does                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Input                                                                           | Output                                          | Tools (EDA)                                                                            | Programs / Skills                                                      |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-------------------------------------------------|----------------------------------------------------------------------------------------|------------------------------------------------------------------------|
|    |                                                                                                                                               | analytically from the routed geometry with a disclosed generic dielectric.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                 |                                                 |                                                                                        |                                                                        |
| 23 |  Post-route STA (multi-corner sign-off)                      | Sign-off timing with real parasitics (MMMC = per-corner loop, one report per corner).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | netlist • SPEF • SDC • multi-corner liberty                                     | post-route timing report • per_corner/          | OpenSTA (per-corner loop)<br>eda_sta                                                   | sta_report_check<br>skills: sta-review                                 |
| 24 |  IR drop (static + dynamic)                                  | Power-grid voltage-drop analysis vs budget: static, plus VCD-vectored dynamic IR (activity-weighted droop under a real switching VCD).                                                                                                                                                                                                                                                                                                                                                                                                                                                    | routed.def (PSM) • VCD                                                          | static + dynamic IR reports                     | OpenROAD PSM ( read_vcd )                                                              | ir_drop_report_check • dynamic_ir_drop_check<br>skills: ir-drop-triage |
| 25 |  EM check (electromigration)                                 | Current-density / metal-lifetime screen.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | routed.def (PSM -enable_em)                                                     | EM report + per-segment currents                | OpenROAD PSM -enable_em                                                                | em_report_check<br>skills: em-check                                    |
| 26 |  Antenna check                                               | Detect + repair process-antenna violations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | routed.def                                                                      | antenna report                                  | OpenROAD check_antennas/repair                                                         | antenna_report_check                                                   |
| 27 |  Signal integrity (crosstalk)                                | Crosstalk/noise screen (SPEF coupling-cap; advisory tier explicitly named).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | SPEF                                                                            | SI report (incl. >0.9 coupling watch-list)      | in-house SPEF coupling screen (OpenSTA window advisory)                                | si_crosstalk_check                                                     |
| 28 |  PERC / Reliability sign-off (ESD + latch-up + cross-domain) | Enforced sign-off: ESD pad-ring + discharge topology, latch-up well-tap, cross-voltage-domain protection; maps to the four PERC categories — netlist checks + netlist-driven layout checks (automated), current density + P2P resistance (named manual-review). Manual review is signed off by a senior physical-design / reliability engineer; criteria live in perc_equivalent.json (categories[],status=MANUAL_REVIEW + review_criteria: PDK Jmax tables, foundry ESD discharge-path P2P limit, Vhold>Vdd, L21 cross-domain contract), results back-filled into checklist[],confirmed. | gate netlist • routed.def • L21 power intent • L3 ESD spec • step-24–27 reports | perc_equivalent.json • PERC memo • gate verdict | in-house PERC-equivalent (DEF-driven)                                                  | perc_signoff_check                                                     |
| 29 | Post-layout gate-level simulation (SDF)                                                                                                       | Gate-level sim with SDF delays to confirm post-layout function (honest SKIP when no SDF re-sim ran).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | gate netlist • SDF • TB                                                         | post-sim results                                | iverilog + SDF<br>eda_simulate                                                         | post_layout_sim_check                                                  |
| 30 | Post-layout SPICE verification                                                                                                                | Transistor-level ngspice correlation vs the Liberty timing: a representative cell plus the top-N STA critical paths, one worst path per distinct endpoint (extracted subckts stitched stage by stage with real net cap; SPICE vs STA path delay per path, aggregated).                                                                                                                                                                                                                                                                                                                    | SPICE decks • SPEF • STA paths                                                  | cell + top-N path SPICE correlation reports     | ngspice + OpenSTA<br>eda_spice                                                         | spice_correlation_check<br>skills: ams-sim                             |
| 31 |  Physical verification (DRC + LVS + ERC + density)         | Sign-off physical rules; density here is <b>RULE COMPLIANCE</b> (KLayout per-layer CMP-window deck; execution verification → Step 34, optimization advisory → Step 35); LVS = Magic extraction + real netgen compare (macro-bearing designs — e.g. Caravel-class harnesses — may blackbox macros: Magic <code>lef write -hide</code> masks the macro to its interface shell, a netgen supplementary setup compares it as a same-name blackbox; the waiver basis = device-level match + KLayout cross-check, written by <code>signoff_waiver_emit</code> into the project's wai            | GDS • gate netlist • PDK decks                                                  | sign-off DRC • LVS • ERC reports                | KLayout DRC • Magic ext2spice + netgen LVS • OpenROAD ERC<br>eda_drc_klayout • eda_lvs | erc_density_check<br>skills: drc-fix • lvs-triage • perc-check         |

| #  | Step                                                                                                | What it does                                                                                                              | Input            | Output                 | Tools (EDA)  | Programs / Skills                  |
|----|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|------------------|------------------------|--------------|------------------------------------|
|    |                                                                                                     | vers.json, cross-referenced as reviewer to-dos by the Step 36 checklist's open_waivers).                                  |                  |                        |              |                                    |
| 32 |  ECO (repair loop) | Engineering-change repair loop when sign-off finds issues (consumes the Step 18 spare cells first, for metal-only fixes). | sign-off reports | ECO log or no-ECO flag | OpenROAD ECO | eco_loop_audit<br>skills: eco-plan |

## Stage 4 — Output & Tapeout

| #  | Step                                                        | What it does                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Input                                    | Output                                                             | Tools (EDA)                                     | Programs / Skills                                                               |
|----|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|--------------------------------------------------------------------|-------------------------------------------------|---------------------------------------------------------------------------------|
| 33 | Power analysis (post-layout)                                | Full-chip power sign-off (post-layout vectorless OpenSTA report_power; optional VCD vector mode).                                                                                                                                                                                                                                                                                                                                                                         | netlist • SDC • liberty (+ optional VCD) | power report (leakage+dynamic, analysis_mode)                      | OpenSTA report_power (+ optional VCD)           | power_report_check<br>skills: power-analysis                                    |
| 34 | Metal fill (density fill insertion)                         | Std-cell-row filler placement (white-space); density here is <b>EXECUTION VERIFICATION</b> (the metal_fill_density_check gate owns the density verdict; rule compliance → Step 31, optimization advisory → Step 35).                                                                                                                                                                                                                                                      | routed.def                               | filled.def • density report                                        | OpenROAD filler_placement<br>eda_pnr            | metal_fill_density_check • spare_cell_preservation_check                        |
| 35 | DFM screen (manufacturability)                              | Manufacturability screen: redundant-via ratio (single-cut fraction advisory) + density <b>OPTIMIZATION ADVISORY</b> (cross-references the Step 34 gate result only — never a duplicate FAIL); OPC/RET/SRAF/PSM as FOUNDRY_SIDE disclosure items (escalated to DESIGNER_COLLAB_REVIEW at ≤28nm; the node is derived by dfm_screen_check from the input/pdk/liberty filenames and recorded in the same dfm_screen.json — process_nm / advanced_node / foundry_side fields). | routed.def • density report              | dfm_screen.json (via stats + foundry-side list)                    | in-house DEF via statistics + density cross-ref | dfm_screen_check                                                                |
| 36 | Tapeout checklist (final sign-off)                          | Item-by-item final confirmation (substance checks: DRC counts, evidence chains).                                                                                                                                                                                                                                                                                                                                                                                          | all sign-off reports                     | tapeout_checklist.json                                             | — (inventory aggregation)                       | tapeout_signoff_check<br>skills: tapeout-checklist                              |
| 37 | GDSII output                                                | Stream the foundry-deliverable GDSII (only when Step 31 PV is fully clean).                                                                                                                                                                                                                                                                                                                                                                                               | routed.def • merged GDS                  | sign-off *.gds                                                     | Magic/KLayout stream-out<br>eda_gds             | gds_size_check • provenance_check                                               |
| 38 | Foundry handoff (mask spec + WAT + scribe + corner vectors) | Foundry physical mask kit: mask spec + WAT plan + scribe PCM + corner ATE vectors (chip-specific; foundry-supplied fields named PENDING_FOUNDRY * — tracked in the Step 36 checklist and back-filled after the foundry replies).                                                                                                                                                                                                                                          | GDS • netlist stats • L10 cases          | mask_spec.json • wat_plan.json • scribe • corner_test_vectors.json | — (pack generator)                              | foundry_handoff_package_check<br>skills: tapeout-checklist                      |
| 39 | FPGA final sign-off (on-board test)                         | Final FPGA recompile + on-board attestation with hardware evidence.                                                                                                                                                                                                                                                                                                                                                                                                       | RTL • board                              | final.sof • on_board_pass.json                                     | Quartus + on-board measurement                  | bringup_plan_gen • fpga_on_board_attestation_check<br>skills: fpga-test-harness |

## Stage 5 — Manufacturing & test (post-fab; triggers on silicon receipt)

| #  | Step                            | What it does                                                            | Input               | Output                         | Tools (EDA)      | Programs / Skills              |
|----|---------------------------------|-------------------------------------------------------------------------|---------------------|--------------------------------|------------------|--------------------------------|
| 40 | Fabrication (foundry, external) | Foundry mask-set + wafer fab (OPC/RET are foundry-side mask synthesis). | foundry handoff kit | mask/wafer intake attestations | external foundry | manufacturing_fab_intake_check |

| #  | Step                                   | What it does                                                                                                                                                                 | Input                         | Output                                       | Tools (EDA)                 | Programs / Skills            |
|----|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------------------------------------|-----------------------------|------------------------------|
| 41 | Wafer sort / probe test                | Wafer probing, good-die selection; yield independently re-derived vs target.                                                                                                 | wafer lot • probe card        | wafer_sort_yield.json • wafer_map.csv        | ATE + probe card (external) | wafer_sort_yield_check       |
| 42 | Packaging (assembly)                   | wirebond / FC-CSP / WLCSP assembly.                                                                                                                                          | good dies                     | packaging_log.json                           | assembly house (external)   | packaging_intake_check       |
| 43 | Final test (ATE + burn-in)             | Post-package final test (functional + parametric + burn-in infant-mortality screen).                                                                                         | packaged units • ATE patterns | final_test_yield.json • burn_in_results.json | ATE (external)              | final_test_attestation_check |
| 44 | Reliability qualification (HTOL / FIT) | Long-duration HTOL qual (device-hours / failures / FIT attestation; required for automotive/medical grades; consumer MPW may stay dormant = DEFERRED, never blocks tapeout). | HTOL chamber results          | htol_results.json verdict                    | HTOL chamber (external)     | htol_attestation_check       |

Out-of-flow lab steps (unnumbered): PFA/EFA (destructive FIB/SEM/EMMI failure analysis) and silicon characterization (shmoo) — data originates from external equipment; the plugin owns the data-analytic root-cause layer (wafer\_map\_pattern\_classify, yield-diagnostic).

## Parallel tracks

### Analog A1–A9 (parallel to Stages 1–3)

| #  | Step                                                                                                                | What it does                                                                                                                                                                | Input                     | Output                                        | Tools (EDA)                                           | Programs / Skills                                                                                                  |
|----|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----------------------------------------------|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| A1 | Analog spec extraction                                                                                              | Extract per-block analog specs from the L docs.                                                                                                                             | L5 ADI spec               | spec.json                                     | deterministic spec extract                            | analog_a1_spec_extract_check<br>skills: analog-spec-extract                                                        |
| A2 | Topology selection                                                                                                  | Choose the circuit topology per spec.                                                                                                                                       | spec.json                 | topology.md                                   | AI topology select                                    | analog_a2_topology_select_check<br>skills: analog-topology-select                                                  |
| A3 | Netlist generation                                                                                                  | Generate the SPICE netlist.                                                                                                                                                 | topology • PDK models     | <block>.sp                                    | xschem netlist<br>eda_xschem_netlist                  | analog_netlist_pdk_check<br>skills: analog-netlist-gen                                                             |
| A4 | Corner sweep (PVT)                                                                                                  | PVT corner sweep + Monte-Carlo yield (mc_yield_pct ≥ 95% gate).                                                                                                             | .sp • PDK corner libs     | corner_results.json (executed/derived counts) | ngspice (corner + MC)<br>eda_spice_corner             | analog_a4_corner_sweep_check<br>skills: analog-sizing-loop                                                         |
| A5 | Analog layout                                                                                                       | Complete the analog layout.                                                                                                                                                 | netlist                   | layout.mag / GDS                              | Magic layout<br>eda_analog_layout                     | analog_a5_layout_check<br>skills: analog-layout                                                                    |
| A6 | Block physical verification (per-block DRC + LVS)                                                                   | Per-block DRC + LVS before merge (catches block-level errors top-level PV would mask).                                                                                      | block GDS • netlist       | DRC clean / LVS match flags                   | Magic DRC + netgen LVS                                | analog_a6_block_pv_check<br>skills: drc-fix • lvs-triage                                                           |
| A7 |  Post-layout resimulation        | Re-simulate with extracted parasitics; compare pre-vs-post specs (>10% degradation loops to A3).                                                                            | layout parasitics • spec  | pre_vs_post.json                              | Magic extraction + ngspice<br>eda_spice_corner        | analog_pre_vs_post_layout_check<br>skills: analog-extraction-resim                                                 |
| A8 | Hardmacro generation (LEF + Liberty + GDS + Verilog)                                                                | Package the block for Stage-3 consumption (LEF should carry obstruction layers — Magic lef write -hide or abstract with obs — so top-level routing cannot enter the macro). | layout • characterisation | 4-file hardmacro                              | Magic/abstract + characterisation                     | analog_hardmacro_check<br>skills: analog-hardmacro-gen                                                             |
| A9 |  Co-simulation / HW verification | Mixed digital+analog simulation and hardware-in-the-loop verification.                                                                                                      | hardmacro • digital RTL   | cosim / HW measurement results                | iverilog + ngspice co-sim<br>eda_spice • eda_simulate | mixed_signal_cosim_check • analog_hw_spice_correlation_check<br>skills: mixed-signal-cosim • analog-hw-tuning-loop |

Mixed-signal M1–M4 (triggered when analog blocks exist)

Trigger timing: runs once **A8 hardmacros are complete** and **Stage 3 is near closure** (routed/GDS mergeable) — hardmacros must exist before floorplan, but M1's GDS merge waits for the digital side to finish routing. If a hardmacro slips until Stage 3 has entered Step 31, M1 waits for the final hardmacro GDS and Step 31's top-level LVS re-runs (incremental — scoped to the macro-interface change).

| #  | Step                                                  | What it does                                                                                                                                                                                                                            | Input                       | Output                                                             | Tools (EDA)                          | Programs / Skills                                                                                                                                         |
|----|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|--------------------------------------------------------------------|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| M1 | Top-level integration (A+D GDS merge)                 | Merge digital + analog GDS, place macros; run top-level LVS on the merged GDS (Magic extraction + netgen, macros blackboxed).                                                                                                           | digital GDS • hardmacro GDS | top_merged.gds • merge/LVS report                                  | KLayout merge + Magic/netgen top-LVS | mixed_signal_merge_check<br>skills: analog-flow-orchestrate • flow-orchestrate                                                                            |
| M2 | Power domain verification (level shifter / isolation) | Structural verification of cross-domain level-shifters / isolation, plus a cross-power-domain signal-crossing check that derives the isolation/level-shifter requirement from the power-domain definitions and audits the UPF strategy. | L21 • merged design         | power_domain / level_shifter / isolation / signal-crossing reports | structural check programs            | power_domain_crossing_check • level_shifter_required_check • isolation_cell_required_check • power_domain_signal_crossing_check<br>skills: ir-drop-triage |
| M3 | Mixed-signal verification (AMS co-sim)                | AMS co-simulation + interface signal integrity.                                                                                                                                                                                         | merged design • cosim TB    | cosim results • interface SI report                                | AMS co-sim                           | mixed_signal_cosim_check • mixed_signal_interface_si_check<br>skills: mixed-signal-cosim • ams-sim                                                        |
| M4 | Mixed-signal sign-off                                 | Top-level verdict rolled up from M1–M3.                                                                                                                                                                                                 | M1–M3 reports               | signoff.json                                                       | rollup verdict                       | mixed_signal_signoff_check<br>skills: tapeout-checklist                                                                                                   |

Totals

| Phase                                                | Stages                                                  | Steps                              |
|------------------------------------------------------|---------------------------------------------------------|------------------------------------|
| Phase 1<br>—<br>Specific<br>ation &<br>docum<br>ents | two entries (Agent • doc-gen) + architecture front-ends | D1–D5 + PM Agent • IC Expert Agent |
| Phase 2<br>— RTL<br>→<br>synthes<br>is               | Stage 1 • Stage 2                                       | 1–14                               |
| Phase 3<br>—<br>Physica<br>l →<br>Tapeou<br>t        | Stage 3 • Stage 4 • Stage 5                             | 15–44                              |
| Parallel                                             | Analog • Mixed-signal                                   | A1–A9 • M1–M4                      |

**44 sequential steps** (Stage 1: 1–6 • Stage 2: 7–14 • Stage 3: 15–32 • Stage 4: 33–39 • Stage 5: 40–44), plus Phase 1 (Agent path & doc-gen path D1–D5) and the two parallel tracks (Analog A1–A9 • Mixed-signal M1–M4). Preflight: P0 (environment health check). Conditional lettered steps: FS1 (ISO-26262 FMEDA diagnostic-coverage, safety designs only) • DT1 (transition-delay-fault ATPG, scan designs only) • DT2 (path-delay-fault at-speed ATPG, scan designs after routing) • DT3 (small-delay-defect at-speed grade, after DT2). Orchestrator `vibe_ic_one_shot_runner.py` runs Phase 1 → Phase 2 → Analog → Phase 3.

Out of scope (declined with reasons): designer-EXECUTED OPC/RET (mask synthesis is foundry-side — surfaced as FOUNDRY\_SIDE items in Step 35 + noted at Step 40), commercial hardware emulators (FPGA path covers the intent), MBIST/LBIST/EDT compression (no open-source engines), BSR/BSDL, automatic clock gating (no characterized ICG cell in sky130; manual RTL clock gating remains available — the SDC declaration of divided/gated clocks is guarded by Step 8's `derived_clock_sdc_required_check`), via-doubling/CAA (commercial DFM).

**E1–E3 external-lab steps (reserved numbering, outside the 44; activated on automotive/medical grade upgrades):**

| #  | Step                                    | Notes                                            |
|----|-----------------------------------------|--------------------------------------------------|
| E1 | PFA / EFA                               | FIB / SEM / EMMI failure analysis (external lab) |
| E2 | Silicon characterization                | shmoo plots • voltage/frequency sweeps           |
| E3 | Temperature cycling / mechanical stress | JESD22-class environmental stress (external lab) |

## Appendix: measured wall-clock reference (for intuition)

Numbers come from REAL local sky130 open-source-flow runs (the `duration_s` fields of `reports/orchestrator/*_one_shot.json`; design sizes 0.5k–21k cells) — measured, never estimated. Durations vary with design, constraints and machine; nonlinear in cell count.

| Stage                                                     | Measured range                                                                                                                                                                                                                            | Samples                                       |
|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| Step 9<br>Synthesis<br>(Yosys)                            | ~0.4 s – 21 s                                                                                                                                                                                                                             | 0.5k cells (lpc) → 20k cells (cv32e40p)       |
| Steps 15–22<br>PnR<br>(OpenROAD, full)                    | ~3 s – 31 min                                                                                                                                                                                                                             | 3.4k cells (subservient) → 21k cells (sha256) |
| Step 37<br>GDS write                                      | ~2 – 4 s                                                                                                                                                                                                                                  | all samples                                   |
| Step 31<br>DRC<br>(KLayout sky130 deck)                   | ~8 – 84 s                                                                                                                                                                                                                                 | 0.5k → 20k cells                              |
| Step 31<br>LVS<br>(Magic + netgen)                        | this batch's samples took the light/skip path, so no representative figure; with real macro compares (e.g. Caravel-class harnesses) it runs <b>minutes to hours</b> , scaling with macro count (a field run motivated the 4-hour timeout) | —                                             |
| Steps 40–43<br>manufacturing<br>(fab/sort/pkg/final test) | external, weeks-scale                                                                                                                                                                                                                     | no local measurement                          |

Licensing & IP: the whole flow relies on open-source tools only (Apache-2.0 project; commercial-tool firewall and output ownership: `repo-root/README.md` § IP ownership, plus the DCO/patent pledge in `CONTRIBUTING.md`).

繁體中文版： `ALL_STEPS.zh-TW.md`.